...d. Linear ...

...t 1

1.2: 2d, 5h

2d. $y' = 2xy^2$; $y(x) = -1/(x^2+A)$, $y(0)=5$

$$y'(x) = \frac{1}{(x^2+A)^2} \cdot 2x = 2x\left(-\frac{1}{(x^2+A)}\right)^2 = 2xy^2$$

$$y(0) = 5 = -1/(0^2+A) = -1/A \rightarrow A = -1/5$$

5h. $y'' + 5yy' + y = 0$. Let $y = e^{rx}$, $y'' + 5yy' + y = r^2e^{rx} + 5e^{rx} \cdot re^{rx} + e^{rx}$

$y'' + 5yy' + y = e^{rx}(r^2 + 5re^{rx} + 1)$. $e^{rx} \neq 0$ for all x and

$(r^2 + 5re^{rx} + 1) \neq 0$. Therefore, $y = e^{rx}$ is not a solution for any r.

2.2: 1f, 2f

1f. $xy' + 3y = 0$.

I. $y(x) = Ae^{-\int p(x)dx}$. $y' + \left(\frac{3}{x}\right)y = 0$, $x \neq 0 \rightarrow p(x) = \frac{3}{x}$

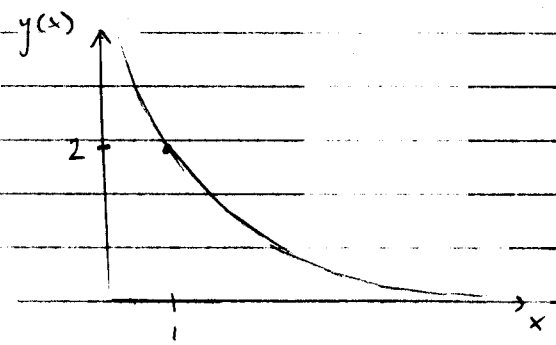
$$y(x) = Ae^{-\int \frac{3}{x}dx} = Ae^{-3\ln x} = Ax^{-3}$$

II. $x\frac{dy}{dx} + 3y = 0 \rightarrow -\frac{3}{x}dx = \frac{dy}{y} \rightarrow \ln y = -3\ln(x)$

$$e^{\ln y} = y = e^c x^{-3} \rightarrow y(x) = Ax^{-3}$$

2f. $y(x) = Ax^{-3}$, $y(1) = 2 \rightarrow A1^{-3} = A = 2$. $y(x) = \frac{2}{x^3}$

The interval is $(0, +\infty)$.



2.3: 1n

$$\frac{dr}{d\theta} + 2(\cot 2\theta) r = 1$$

I $r(\theta) = e^{-\int p(\theta)d\theta}\left(\int e^{\int p(\theta)d\theta} q(\theta)\,d\theta + c\right)$

$p(\theta) = 2\cot 2\theta$, $q(\theta) = 1$

$r(\theta) = e^{-\int 2\cot 2\theta\, d\theta}\left(\int e^{\int 2\cot 2\theta\, d\theta}\, d\theta + c\right)$

$e^{2\int \cot 2\theta\, d\theta} = \sin 2\theta$.

$r(\theta) = (\sin 2\theta)^{-1}\left(-\frac{\cos 2\theta}{2} + c\right) = -\frac{\cot(2\theta)}{2} + \frac{C}{\sin 2\theta}$

II $\sigma(\theta) = \sin 2\theta$, $\sigma(\theta)\frac{dr}{d\theta} + 2\cot 2\theta \sin 2\theta\, r = \sigma(\theta)$

$\sin 2\theta \frac{dr}{d\theta} + 2\cos 2\theta\, r = \sigma(\theta) = \frac{d}{d\theta}\left(\sin 2\theta \cdot r(\theta)\right) = \sin 2\theta$

$\sin 2\theta\, r(\theta) = -\frac{1}{2}\cos 2\theta + C \rightarrow r(\theta) = -\frac{\cot(2\theta)}{2} + \frac{C}{\sin(2\theta)}$

Set 2

3.2: 1K, 5b, 6c

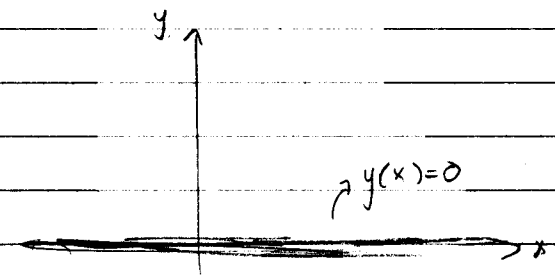
1K. $y' + 3y(y+1)\sin 2x = 0$, $y(0) = 0$

$\frac{y'}{3y(y+1)} = -\sin 2x$, $\frac{1}{y(y+1)} = \frac{1}{y} - \frac{1}{y+1}$ by partial fraction decomp.

$\int \frac{dy}{y} - \int \frac{dy}{y+1} = -3\int \sin(2x)\,dx = \frac{3}{2}\cos(2x) + c = \ln y - \ln(y+1)$

$\ln\left(\frac{y}{y+1}\right) = \frac{3}{2}\cos(2x) + c$, $\frac{y}{y+1} = e^c \cdot e^{\frac{3}{2}\cos(2x)} = A e^{\frac{3}{2}\cos(2x)}$

$\frac{y(0)}{y(0)+1} = 0 = Ae^{3/2} \rightarrow A = 0$. Therefore, $y(x) = 0$ is the solution



5b $y' = f(y/x)$

$w = y/x \rightarrow \frac{d}{dx} w(x) = \frac{xy' - y}{x^2}$ by the Chain rule.

$w'(x) = \frac{y' - y/x}{x} = \frac{f(y/x) - y/x}{x} = \frac{f(w) - w}{x}$

6e $y' = \sqrt{1 + (y/x)^2} + (y/x)$. Let $w(x) = y/x$, $y' = f(y/x) \rightarrow w' = \frac{f(w) - w}{x}$

$w' = \frac{\sqrt{1+w^2} + w - w}{x} = \frac{\sqrt{1+w^2}}{x}$; $\frac{dw}{\sqrt{1+w^2}} = \frac{dx}{x} \rightarrow \int \frac{dw}{\sqrt{1+w^2}} = \ln\left(\left|\sqrt{w^2+1} + w\right|\right) = \ln|x| + c$

or $\sinh^{-1}(w) = \ln|x| + c \rightarrow w = \frac{y}{x} = \sinh(\ln|x| + C) \rightarrow y(x) = x\sinh(\ln|x| + c)$

3.3: 1b $y' = x + y$, $y(0) = 3$, $y(0) = 0$, $y(0) = -2$, $y(-1) = 0$

Both $f(x,y) = x + y$, $f_y(x,y) = 1$ are continuous for all x, y so Theorem 3.3.1 assures the existence of a unique solution.

(Theorem 2.3.1 predicts the solution to exist on $-\infty < x < \infty$).

The solutions are:

$y(x) = 4e^x - x - 1$, $y(0) = 3$

$y(x) = e^x - x - 1$, $y(0) = 0$

$y(x) = -e^{-x} - x - 1$, $y(0) = -2$

$y(x) = -x - 1$, $y(-1) = 0$.

$\frac{dy}{dx} = x + y \rightarrow \frac{dy}{dx} - y = x \rightarrow y(x) = e^{\int dx}\left(\int e^{-\int dx} x\, dx + C\right)$

$y(x) = e^x\left(\int e^{-x} x\, dx + C\right) = Ce^x + e^x(-e^{-x}(x+1)) = Ce^x - (x+1)$

Set 3

3.5: 1b. $N' - aN = -bN^2$, $v = N^{1-2} = N^{-1} \rightarrow v' + (1-2)(-a)v = (1-2)(-b)$

or $v' + av = b \rightarrow v(t) = Ae^{-at} + b/a$ or $N(t) = \frac{a}{Be^{-at} + b}$

10.1 2b

Use Kirchhoff's Laws

Assume $i_1 > i_2$

Loop 1: $-i_1 R - \frac{1}{C}\int i_1\,dt - E(t)$

Loop 2: $+E(t) - \frac{1}{C}\int i_2\,dt - L\frac{di_2}{dt}$

Taking derivatives; 1. $-E'(t) = i_1' R + \frac{1}{C} i_1$, 2. $E'(t) = \frac{1}{C} i_2 + L i_2''$

10.2 2b

$x_1' = x_3 + 8$

$x_2' = x_3 + 8t$

$x_3' = x_1 + x_2 + x_3$

Check x_1: With $x_1(t) = 1, x_2(t) = -1, x_3(t) = 0$

$x_1' - x_3 = 0 - 0 = 0$ ✓

$x_2' - x_3 = 0 - 0 = 0$ ✓

$x_3' - x_1 - x_2 - x_3 = 0 - 1 + 1 - 0 = 0$ ✓

Check x_2: $x_1(t) = e^{-t}, x_2(t) = e^{-t}, x_3(t) = -e^{-t}$,

$x_1' - x_3 = -e^{-t} + e^{-t} = 0$ ✓

$x_2' - x_3 = -e^{-t} + e^{t} = 0$ ✓

$x_3' - x_1 - x_2 - x_3 = e^{-t} - e^{-t} - e^{-t} + e^{-t} = 0$ ✓

Check x_3: $x_1(t) = e^{2t} + 3, x_2(t) = e^{2t} - 3, x_3(t) = 2e^{2t}$

$x_1' - x_3 = 2e^{2t} - 2e^{2t} = 0$ ✓

$x_2' - x_3 = 2e^{2t} - 2e^{2t} = 0$ ✓

$x_3' - x_1 - x_2 - x_3 = 4e^{2t} - e^{2t} - 3 - e^{2t} + 3 - 2e^{2t} = 0$ ✓

Check x_p: $x_{1p}(t) = -2t^2 + 6t - 1, x_{2p}(t) = 2t^2 - 2t - 1, x_{3p}(t) = -4t - 2$

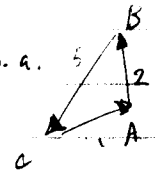
$x_1' - x_3 - 8 = -4t + 6 + 4t + 2 - 8 = 0$ ✓

$x_2' - x_3 - 8t = 4t - 2 + 4t + 2 - 8t = 0$ ✓

$x_3' - x_1 - x_2 - x_3 = -4 + 2t^2 - 6t + 1 - 2t^2 + 2t + 1 + 4t + 2 = 0$ ✓

General solution: $\vec{x}(t) = C_1\begin{pmatrix}1\\-1\\0\end{pmatrix} + C_2\begin{pmatrix}e^{-t}\\e^{-t}\\-e^{-t}\end{pmatrix} + C_3\begin{pmatrix}e^{2t}+3\\e^{2t}-3\\2e^{2t}\end{pmatrix} + \begin{pmatrix}-2t^2+6t-1\\2t^2-2t-1\\-4t-2\end{pmatrix}$

With $C_1 = -123$, $C_2 = -40$, $C_3 = 41$

3. a. 

Law of [illegible] $5^2 = 1^2 + [illegible] 2(1)(2)\cos\theta$ or [illegible]

3b. Law of Cosines: $5^2 = 4^2 + 2^2 - 2(4)(2)\cos\theta \Rightarrow \cos\theta$

a solution

4.3 1h Undefined since $\vec{t}$, $\vec{u}$ are 4-tuples, $\vec{v}$ is a 3-tuple